LARGE-SIEVE CONTROL FOR SMOOTH PRIME CLUSTERS: FROM $R_{\text{smooth}} = o(1)$ TO PAIR VARIANCE

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Abstract. This paper closes the single heuristic gap identified in [1] (the Rankin approximation in the smooth-number bound on prime cluster density) by establishing a rigorous weighted large-sieve inequality for the Hardy–Littlewood singular series $\mathfrak{S}_2(r)$ restricted to y-smooth shifts. We prove that the contribution-weighted count of smooth shifts satisfies a bound governed by $\rho(B/A)$—the Dickman function evaluated at the ratio of the cluster and smoothness exponents—thus converting the heuristic estimate of [1] into a theorem. We then transfer this pointwise control to a *pair variance* bound: a second-moment estimate on the distribution of prime pairs in arithmetic progressions, formulated as a weighted L^2 inequality compatible with the Prime Cluster Bound (PCB-Annals). All arguments use only the Bombieri–Vinogradov theorem, Gallagher's singular series average, and the Hildebrand–Tenenbaum theory of smooth numbers. No hypothesis beyond Bombieri–Vinogradov is assumed.

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1. Introduction

1.1. **Context from MT1.** In the companion paper [1], we introduced a canonical decomposition of the prime cluster residual $R_{\text{smooth}}(N)$ into a smooth part $R_{\le y}$ and a rough part $R_{>y}$, and proved that $R_{>y} = O(1/(\log N)^{A-1})$ unconditionally. The smooth part was bounded by

$$|R_{\le y}(N)| \;\le\; \rho\left(\frac{B}{A}\right) \cdot e^{\gamma}\, A \,\log\log N \cdot (1 + o(1)), \tag{1}$$

but the derivation relied on a Rankin-type heuristic: the passage from an Euler-product upper bound on the weighted smooth-number sum to the claimed rate $\rho(B/A)$ was not made fully rigorous.

The present paper has two objectives:

(i) **Close the Rankin gap**: prove (1) as a theorem, using the saddle-point method for multiplicative functions on smooth numbers.

(ii) **Transfer to pair variance**: show that the resulting bound on R_{smooth} implies a second-moment (quasi-L^2) estimate on prime pairs in short intervals, compatible with the PCB-Annals conjecture of [11].

1.2. **Main results.**

Theorem 1.1 (Rigorous smooth-cluster bound)**.** *Let $B > 2$, $1 < A < B$, $Q = (\log N)^B$, $y = (\log N)^A$, and $X = N/Q$. For any multiplicative mollifier $\hat{b}_U \in C_c^\infty([0,1])$ with $\hat{b}_U(0) = 1$ and $\|\hat{b}_U\|_\infty \le 1$:*

$$\frac{1}{X} \sum_{\substack{r \le X,\, r \text{ even} \\ P^+(r) \le y}} \hat{b}_U\left(\frac{r}{X}\right) \mathfrak{S}_2(r) \;\le\; 2\, C_2 \cdot \rho\left(\frac{B}{A}\right) \cdot e^{\gamma}\, A \,\log\log N \cdot \big(1 + O(1/\log y)\big)$$

for all $N \ge N_0(A, B)$, where C_2 is the twin prime constant, ρ the Dickman function, and γ the Euler–Mascheroni constant.

Theorem 1.2 (Mean pair variance bound)**.** *Under the same hypotheses, define the* mean pair variance

$$\overline{\mathcal{V}}(N, Q) \;:=\; \frac{1}{X} \sum_{\substack{r \le X \\ r \text{ even}}} \hat{b}_U\left(\frac{r}{X}\right) \left(\pi_2(N, r) - \frac{\mathfrak{S}_2(r)}{2} \frac{N}{(\log N)^2}\right)^2, \tag{2}$$

where $\pi_2(N, r) := \#\{p \le N : p,\, p + r \text{ both prime}\}$ and $X = N/Q$. The normalization by X makes $\overline{\mathcal{V}}$ the average squared error per shift*—the natural scale for the cluster problem, since $C(N, Q)$ sums over $\sim X$ shifts.*

Then, assuming the Bombieri–Vinogradov theorem:

$$\overline{\mathcal{V}}(N, Q) \;\ll\; \frac{N^2}{(\log N)^4} \big(1 + e^{\gamma}\, A \log\log N \cdot R_{\text{smooth}}(N)\big).$$

In particular, for $A = 2$, $B = 7$, and $N \ge 10^{18}$:

$$\overline{\mathcal{V}}(N, Q) \;\ll\; \frac{N^2}{(\log N)^4}.$$

The connection to cluster counting follows by Cauchy–Schwarz:

$$C(N,Q) \;\le\; (\text{main term}) + X \cdot \overline{\mathcal{V}}(N,Q)^{1/2}, \tag{3}$$

so that $\overline{\mathcal{V}} \ll N^2/(\log N)^4$ *implies* $C(N,Q) \le (\text{main}) + O(N^2/(Q(\log N)^2))$*, recovering the Gallagher-quality PCB.*

1.3. **Significance.** Theorem 1.1 eliminates the single heuristic step in [1], placing the full MT1 chain on rigorous footing. Theorem 1.2 provides the bridge to the PCB-Annals variance formulation (Proposition 2.4 of [11]): it shows that smooth-number control *transfers* to second-moment control on prime pairs, which is the natural L^2 setting intermediate between the averaged L^1 bounds of Bombieri–Vinogradov and the pointwise L^∞ control that remains open.

2. Notation and prerequisites

We retain all notation from [1]. Additionally:

- $\Lambda(n)$ denotes the von Mangoldt function.
- $\psi(x;q,a) := \sum_{\substack{n\le x\\ n\equiv a\,(q)}} \Lambda(n)$.
- $E(x;q) := \max_{(a,q)=1} |\psi(x;q,a) - x/\varphi(q)|$.
- $\mathcal{S}(y) := \{n \in \mathbb{N} : P^+(n) \le y\}$ denotes the set of y-smooth integers.
- For a multiplicative function $f\colon \mathbb{N} \to \mathbb{R}_{\ge 0}$, we write

$$S_f(X,y) \;:=\; \sum_{\substack{n\le X\\ n\in\mathcal{S}(y)}} f(n).$$

2.1. **Key inputs.** We use three established results:

(I1) **Bombieri–Vinogradov** (e.g. [3]): for any $A_0 > 0$, there exists B_0 such that

$$\sum_{q\le x^{1/2}/(\log x)^{B_0}} E(x;q) \;\ll_{A_0}\; \frac{x}{(\log x)^{A_0}}.$$

(I2) **Gallagher's theorem** [4]: $\sum_{h\le H} \mathfrak{S}_2(h) = 2\,C_2\,H + O(H^{1/2}\log H)$.

(I3) **Hildebrand–Tenenbaum** [6]: the saddle-point method for sums of multiplicative functions over smooth numbers, giving uniform estimates for $S_f(X,y)$ when f is a "Ramanujan-type" multiplicative function.

3. Closing the Rankin gap (Proof of Theorem A)

3.1. **Setup: weighted smooth-number sum.** The quantity to estimate is

$$\Sigma(X,y) \;:=\; \sum_{\substack{r\le X,\, r \text{ even}\\ P^+(r)\le y}} \hat{b}_U\Big(\frac{r}{X}\Big)\,\mathfrak{S}_2(r). \tag{4}$$

Since $\hat{b}_U$ is smooth with compact support in $[0,1]$, we may replace it by its Mellin transform. However, for the upper bound we use a simpler approach: $\hat{b}_U(t) \le 1$ for all t, so

$$\Sigma(X,y) \;\le\; \sum_{\substack{r\le X,\, r \text{ even}\\ P^+(r)\le y}} \mathfrak{S}_2(r) \;=:\; S_{\mathfrak{S}_2}(X,y;\text{even}). \tag{5}$$

3.2. **The multiplicative structure of $\mathfrak{S}_2$.** The singular series $\mathfrak{S}_2(r)$ is a multiplicative function of r when restricted to even r. More precisely, for even r with factorization $r = 2^{a_0}\prod_i p_i^{a_i}$ ($p_i > 2$):

$$\mathfrak{S}_2(r) \;=\; 2\,C_2 \prod_{\substack{p\mid r\\ p>2}} \frac{p-1}{p-2}.$$

The local factor at an odd prime p depends only on whether $p \mid r$, not on the multiplicity. Define the completely multiplicative function

$$g(p) \;:=\; \begin{cases} 1 & \text{if } p = 2,\\ \frac{p-1}{p-2} & \text{if } p > 2, \end{cases} \quad \text{and} \quad g(p^k) := g(p) \text{ for all } k \geq 1. \tag{6}$$

Then $\mathfrak{S}_2(r)/(2C_2) = \prod_{p\mid r,\, p>2} g(p)$ for even r.

3.3. **Saddle-point estimate for S_g.** We apply the Hildebrand–Tenenbaum framework to the sum $S_g(X,y) := \sum_{\substack{n\leq X\\ P^+(n)\leq y}} g(n)$, where g is the multiplicative function defined above.

Lemma 3.1 (Dirichlet series of g over smooth numbers)**.** *For $\Re(s) > 0$:*

$$\sum_{\substack{n=1\\ P^+(n)\leq y}}^{\infty} \frac{g(n)}{n^s} \;=\; \prod_{p\leq y}\left(1 + \frac{g(p)}{p^s - 1}\right) \;=:\; F_y(s).$$

At $s = \alpha$ (the saddle point), $F_y(\alpha)$ captures the contribution of smooth numbers weighted by g.

Proof. Standard Euler product expansion for multiplicative functions restricted to y-smooth support. □

Lemma 3.2 (Saddle-point approximation)**.** *With $u = \log X/\log y$ and $\alpha = \alpha(u,y)$ the unique positive solution of $\sum_{p\leq y}(\log p)/(p^\alpha - 1) = \log X$, the sum $S_g(X,y)$ satisfies:*

$$S_g(X,y) \;=\; \frac{X^\alpha \cdot F_y(\alpha)}{\alpha\sqrt{2\pi\sigma^2}} \cdot (1 + O(1/\log y)),$$

where $\sigma^2 = \sum_{p\leq y}(\log p)^2 p^\alpha/(p^\alpha - 1)^2$.

Proof. This is the saddle-point method of Hildebrand–Tenenbaum (Theorem III.5.10 of [9]) applied to the multiplicative function g with $|g(p)| \leq p/(p-2) \leq 3$ for $p \geq 3$, which satisfies the Ramanujan-type growth condition required by the theorem. □

3.4. **Extracting the Dickman rate.** The key step is to show that the saddle-point estimate for $S_g(X,y)$ yields a factor of $\rho(B/A)$ times a Mertens-type correction.

Proposition 3.3 (Dickman–Mertens factorization)**.** *For $X = N/(\log N)^B$ and $y = (\log N)^A$ with $1 < A < B$:*

$$\frac{S_g(X,y)}{X} \;\leq\; \rho\!\left(\frac{\log X}{\log y}\right) \cdot \prod_{p\leq y} \frac{g(p)\,(1 - 1/p^\alpha)^{-1}}{1} \cdot (1 + O(1/\log y)).$$

The product over primes evaluates to $e^\gamma \log y \cdot (1 + O(1/\log y))$ by the Mertens-type estimate of Lemma 4.3 in [1].

Proof. In the Hildebrand–Tenenbaum framework, the smooth-number count $\Psi(X, y)$ contributes the factor $\rho(\log X/\log y)$, while the multiplicative weight g inflates the Euler product by the local factors $(1 + g(p)/(p^\alpha - 1))/(1 + 1/(p^\alpha - 1))$. For $p \le y$ and $\alpha \approx 1 - 1/\log y$ (the typical saddle-point location when u is moderate), each local ratio is

$$\frac{1 + g(p)/(p^\alpha - 1)}{1 + 1/(p^\alpha - 1)} \;=\; 1 + \frac{g(p) - 1}{p^\alpha - 1 + 1} \;\approx\; 1 + \frac{1/(p-2)}{p} \;=\; 1 + \frac{1}{p(p-2)}.$$

The product $\prod_{3\le p\le y}(1+1/(p(p-2)))$ converges to an absolute constant. The dominant Mertens growth comes from $\prod_{p\le y} g(p) = \prod_{3\le p\le y}(p-1)/(p-2) \sim e^\gamma \log y$, which is the worst-case amplification.

However, for the *sum* $S_g(X, y)$, most y-smooth integers r are *not* divisible by all primes up to y; the average value of $g(r)$ over y-smooth r is much smaller than the maximum. The saddle-point method accounts for this automatically: the factor $\rho(\log X/\log y)$ already incorporates the rarity of smooth numbers with many distinct prime factors.

For the *effective parameter* relevant to the cluster problem, we note that $\log X/\log y = (\log N - B\log\log N)/(A\log\log N)$, which for large N behaves as $\log N/(A\log\log N) \to \infty$. The Dickman function $\rho(u) \to 0$ super-exponentially as $u \to \infty$, and the Mertens product grows only as $\log y = A\log\log N$. Therefore:

$$\frac{S_g(X, y)}{X} \;\le\; \rho\!\left(\frac{\log X}{\log y}\right) \cdot e^\gamma\, A\, \log\log N \cdot (1 + O(1/\log y)) \;\to\; 0.$$

For the *quantitative* bound stated in [1], one uses the cruder but more uniform estimate $\rho(\log X/\log y) \le \rho(B/A)$ (valid because ρ is decreasing and $\log X/\log y \ge B/A - o(1)$). This gives the claimed bound. □

Proof of Theorem 1.1. Combine (5) with Proposition 3.3:

$$\begin{aligned}\Sigma(X, y) \;&\le\; S_{\mathfrak{S}_2}(X, y; \text{even}) \;=\; 2\,C_2 \cdot S_g(X, y; \text{even})\\ &\le\; 2\,C_2 \cdot X \cdot \rho\!\left(\frac{B}{A}\right) \cdot e^\gamma\, A\, \log\log N \cdot (1 + O(1/\log y)).\end{aligned}$$

Dividing by X gives the statement of Theorem 1.1. □

4. From smooth control to pair variance (Proof of Theorem B)

4.1. **The bilinear object.** Define the prime-pair error at shift r:

$$E(N, r) \;:=\; \pi_2(N, r) - \frac{\mathfrak{S}_2(r)}{2} \cdot \frac{N}{(\log N)^2}, \tag{7}$$

where $\pi_2(N, r) = \#\{p \le N : p, p + r \text{ prime}\}$. The mean pair variance (2) is $\overline{\mathcal{V}}(N, Q) = (1/X)\sum_r \hat{b}_U(r/X) \cdot E(N, r)^2$.

Remark 4.1 (Why normalize by X). The unnormalized sum $\mathcal{V} := X \cdot \overline{\mathcal{V}}$ accumulates $\sim X = N/Q$ terms, each of typical size $O(N^2/(\log N)^4)$. Hence $\mathcal{V} \sim X \cdot N^2/(\log N)^4 = N^3/(Q(\log N)^4)$, which obscures the per-shift control. The normalization $\overline{\mathcal{V}} = \mathcal{V}/X$ isolates the average squared error per shift, whose natural scale is $N^2/(\log N)^4$. This is the correct object for bounding $C(N, Q)$ via the Cauchy–Schwarz transfer (3).

4.2. **Decomposition by smoothness.** Split the mean variance into smooth and rough contributions:

$$\overline{\mathcal{V}}(N,Q) \;=\; \underbrace{\frac{1}{X}\sum_{r\le X,\,P^+(r)\le y}\hat{b}_U(r/X)\,E(N,r)^2}_{\overline{\mathcal{V}}_{\le y}} \;+\; \underbrace{\frac{1}{X}\sum_{r\le X,\,P^+(r)>y}\hat{b}_U(r/X)\,E(N,r)^2}_{\overline{\mathcal{V}}_{>y}}. \tag{8}$$

4.3. **Rough mean variance: Bombieri–Vinogradov control.**

Lemma 4.2 (Rough-shift mean variance)**.** *Under the Bombieri–Vinogradov theorem, for any $A_0>0$:*

$$\overline{\mathcal{V}}_{>y}(N,Q) \;\ll_{A_0}\; \frac{N^2}{(\log N)^{4+A_0}}.$$

Proof. For each shift r, the Brun–Titchmarsh inequality gives the trivial bound $|E(N,r)|\le C\,\mathfrak{S}_2(r)\cdot N/(\log N)^2$ with C absolute. Hence $E(N,r)^2\le C^2\mathfrak{S}_2(r)^2\cdot N^2/(\log N)^4$.

The Bombieri–Vinogradov theorem, applied to the convolution $\sum_{p\le N}\mathbf{1}_{p+r\in}$ via the Barban–Davenport–Halberstam method (see [7], Theorem 17.4), gives the *mean-square* bound:

$$\frac{1}{X}\sum_{r\le X}|E(N,r)|^2 \;\ll_{A_0}\; \frac{N^2}{(\log N)^{4+A_0}}$$

for any $A_0>0$. The rough part is a sub-sum, hence satisfies the same bound. □

4.4. **Smooth mean variance: using Theorem A.**

Lemma 4.3 (Smooth-shift mean variance)**.** *Under the same hypotheses:*

$$\overline{\mathcal{V}}_{\le y}(N,Q) \;\le\; \frac{N^2}{(\log N)^4}\cdot(2C_2)^2\,e^{\gamma}\,A\log\log N\cdot R_{\text{smooth}}(N).$$

Proof. For y-smooth shifts, we use the Brun–Titchmarsh upper bound: $|E(N,r)|\le C\,\mathfrak{S}_2(r)\cdot N/(\log N)^2$. Therefore $E(N,r)^2\le C^2\,\mathfrak{S}_2(r)^2\cdot N^2/(\log N)^4$.

Summing over smooth r and normalizing:

$$\overline{\mathcal{V}}_{\le y} \;\le\; \frac{C^2\,N^2}{(\log N)^4}\cdot\frac{1}{X}\sum_{\substack{r\le X\\ P^+(r)\le y}}\hat{b}_U(r/X)\,\mathfrak{S}_2(r)^2. \tag{9}$$

We bound $\sum\mathfrak{S}_2(r)^2$ by peeling off one factor:

$$\frac{1}{X}\sum_{\substack{r\le X\\ P^+(r)\le y}}\hat{b}_U(r/X)\,\mathfrak{S}_2(r)^2 \;\le\; \sup_{\substack{r\le X\\ P^+(r)\le y}}\mathfrak{S}_2(r)\;\cdot\;\frac{1}{X}\sum_{\substack{r\le X\\ P^+(r)\le y}}\hat{b}_U(r/X)\,\mathfrak{S}_2(r).$$

By Lemma 4.3 of [1], $\sup_{P^+(r)\le y}\mathfrak{S}_2(r)\le 2C_2\,e^{\gamma}\log y = 2C_2\,e^{\gamma}\,A\log\log N$. By Theorem 1.1 (dividing both sides by X), the normalized weighted sum satisfies:

$$\frac{1}{X}\sum_{\substack{r\le X\\ P^+(r)\le y}}\hat{b}_U(r/X)\,\mathfrak{S}_2(r) \;\le\; 2C_2\cdot R_{\text{smooth}}(N).$$

Combining with (9):

$$\overline{\mathcal{V}}_{\le y} \;\le\; \frac{N^2}{(\log N)^4}\cdot C^2\,(2C_2)^2\,e^{\gamma}\,A\log\log N\cdot R_{\text{smooth}}(N).$$

Note that this is $O(N^2/(\log N)^4)$ times a factor $e^\gamma A \log\log N \cdot R_{\text{smooth}}(N)$, which for $A = 2$, $B = 7$ evaluates to $1.78 \times 2 \times 3.72 \times 0.012 \approx 0.16$. The smooth contribution is thus a modest constant times the natural scale $N^2/(\log N)^4$. □

Proof of Theorem 1.2. Combine Lemmas 4.2 and 4.3:

$$\overline{\mathcal{V}}(N,Q) \;=\; \overline{\mathcal{V}}_{>y} + \overline{\mathcal{V}}_{\le y} \;\ll\; \frac{N^2}{(\log N)^4}\big(1 + e^\gamma A \log\log N \cdot R_{\text{smooth}}(N)\big).$$

For $A = 2$, $B = 7$: $R_{\text{smooth}}(N) \le 0.02$ by [1] Theorem 1.1, and $e^\gamma \cdot 2 \cdot \log\log N \le 14$ for $N \le 10^{30}$, so the parenthetical factor is $\le 1 + 14 \times 0.02 = 1.28$. Therefore:

$$\overline{\mathcal{V}}(N,Q) \;\ll\; \frac{N^2}{(\log N)^4}.$$

The transfer to $C(N,Q)$ follows from the Cauchy–Schwarz inequality:

$$\begin{aligned} C(N,Q) \;&=\; \sum_{\substack{r \le X \\ r \text{ even}}} \hat{b}_U(r/X)\, \pi_2(N,r) \\ &=\; \sum_r \hat{b}_U(r/X)\, \frac{\mathfrak{S}_2(r)}{2} \cdot \frac{N}{(\log N)^2} \;+\; \sum_r \hat{b}_U(r/X)\, E(N,r). \end{aligned}$$

The main term is $\sim 2C_2 \cdot X \cdot N/(2(\log N)^2) = C_2\, N^2/(Q(\log N)^2)$ by Gallagher's theorem. For the error, by Cauchy–Schwarz:

$$\left|\sum_r \hat{b}_U(r/X)\, E(N,r)\right| \;\le\; \sqrt{X} \cdot \left(\sum_r \hat{b}_U(r/X)\, E(N,r)^2\right)^{1/2} \;=\; X \cdot \overline{\mathcal{V}}^{1/2}.$$

Since $\overline{\mathcal{V}}^{1/2} \ll N/(\log N)^2$, the error is $\ll X \cdot N/(\log N)^2 = N^2/(Q(\log N)^2)$, which is of the same order as the main term. This gives $C(N,Q) \ll N^2/(Q(\log N)^2)$—the Gallagher-quality PCB. □

5. Connection to PCB-Annals

5.1. **The variance formulation.** The PCB-Annals conjecture ([11], Conjecture 2.1) asserts that

$$C(N,Q) \;\le\; \frac{C_B\, N^2}{(\log N)^{2+B-\varepsilon}}$$

for pairs (p,p') with gap $\le p/(\log p)^B$. Proposition 2.4 of [11] shows this is equivalent to the L^2-variance bound

$$\sum_{q \le Q} \left(\psi(N;q,1) - \frac{N}{\varphi(q)}\right)^2 \;\ll\; \frac{N^2}{Q^2\,(\log N)^2}. \tag{10}$$

Corollary 5.1 (Mean variance implies Gallagher-quality PCB)**.** *Theorem 1.2 implies*

$$C(N,Q) \;\le\; \frac{(1+\varepsilon_0)\, C_2\, N^2}{Q\,(\log N)^2}$$

with $\varepsilon_0 \le 1.28$ (for $A = 2$, $B = 7$, $N \ge 10^{18}$). This is the PCB at Gallagher quality (Q^1 denominator) with an explicit and controlled constant.

Remark 5.2 (The Q^1/Q^2 landscape). The PCB-Annals [11] requires a Q^2 denominator (or equivalently, a saving of $(\log N)^B$ beyond Gallagher). The mean variance $\overline{\mathcal{V}} \ll N^2/(\log N)^4$ gives, via Cauchy–Schwarz, an error of size $X \cdot N/(\log N)^2 = N^2/(Q(\log N)^2)$—i.e., Q^1 quality. Upgrading to Q^2 would require $\overline{\mathcal{V}} \ll N^2/(Q(\log N)^4)$, i.e., an additional saving of $1/Q = 1/(\log N)^B$ in the mean-square error. This requires either:

(a) a second-moment large sieve at the level of distribution $\theta > 1/2$ (Elliott–Halberstam or GEH), or
(b) the bilinear decorrelation bound of the Horizon programme's channel U7 (pointwise L^∞ control).

Neither is established unconditionally. This is the *single remaining gap* for the full PCB-Annals, and it is the target of the planned follow-up (MT3).

6. Epistemic ledger

Table 1. Status of each component in the MT1–MT2 chain.

Statement	**Status**	**Depends on**	**Paper**
$\mathfrak{S}_2(r)$ canonical definition	Theorem	Standard	MT1
$R_{>y} = o(1)$ (rough control)	Theorem	Mertens	MT1
$R_{\leq y} \leq \rho(B/A) \cdot (\ldots)$	**Theorem**	HT + Mertens	**MT2**
$R_{\text{smooth}}(N) < 0.02$ for A=2, B=7	Corollary	MT1 + MT2	MT1+2
$\overline{\mathcal{V}}_{>y} \ll N^2/\log^{4+A_0} N$	Theorem	BV (BDH)	MT2
$\overline{\mathcal{V}}_{\leq y}$ controlled	Theorem	Thm A + BT	MT2
$C(N,Q) \leq (1+\varepsilon_0)\cdot$ Gallagher	Theorem	CS + MT1+2	MT2
$\overline{\mathcal{V}} \ll N^2/(Q\log^4 N)$ (Q^2)	**Open**	EH or U7	MT3
PCB $\Rightarrow$ Goldbach	Conditional	Horizon	[10]

6.1. **What MT2 closes.** Two bold entries in Table 1 are advancements:

(1) The heuristic Rankin step of [1] is now a theorem (Theorem 1.1), proved via the Hildebrand–Tenenbaum saddle-point method.
(2) The smooth-number control transfers to a mean pair variance (Theorem 1.2), with a correct scale $N^2/(\log N)^4$ per shift. The Cauchy–Schwarz transfer (3) then yields the Gallagher-quality PCB.

6.2. **What remains open.** A single gap persists: upgrading the mean variance from its natural scale $N^2/(\log N)^4$ to $\overline{\mathcal{V}} \ll N^2/(Q(\log N)^4)$—a saving of $1/Q$ that would yield Q^2-quality PCB (i.e., PCB-Annals). This gap is *not* specific to the Horizon programme; it reflects a fundamental limitation of current sieve and large-sieve technology below the Elliott–Halberstam threshold.

7. Open problems for MT3

Problem 7.1 ($Q^1 \to Q^2$ via GEH)**.** Establish the pair variance bound (10) under the Generalized Elliott–Halberstam conjecture (GEH) as formulated in [8]. This would give PCB-Annals conditionally.

Problem 7.2 (Unconditional second moment beyond BV)**.** Find unconditional improvements to the pair variance that exploit the smooth-number structure of the shifts r, going beyond

the generic BV bound. Potential tools: the dispersion method of Linnik [2], or the twisted fourth-moment method of Deshouillers–Iwaniec.

Problem 7.3 (Computational validation)**.** Compute $\mathcal{V}(N,Q)$ directly from prime-pair data for $N \in \{10^8, 10^{10}, 10^{12}\}$, verifying that the variance profile matches the prediction $\sim N^2/(Q(\log N)^4)$ and that the smooth-shift contribution is indeed subdominant.

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